

Case Analysis – Avoiding Misleading Proportions

Activity 6

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Question 1: Truncated Y-Axis in Sales Growth

A retail company's monthly sales data (in ₹ lakhs) is as follows:

Month	Sales (₹ lakhs)
January	95
February	97
March	99
April	101
May	103
June	105

The marketing team's chart begins the Y-axis at 90 instead of 0, which makes the increase from January to June look almost like a doubling of sales, when in fact it is only a modest rise.

1. Why the chart is misleading

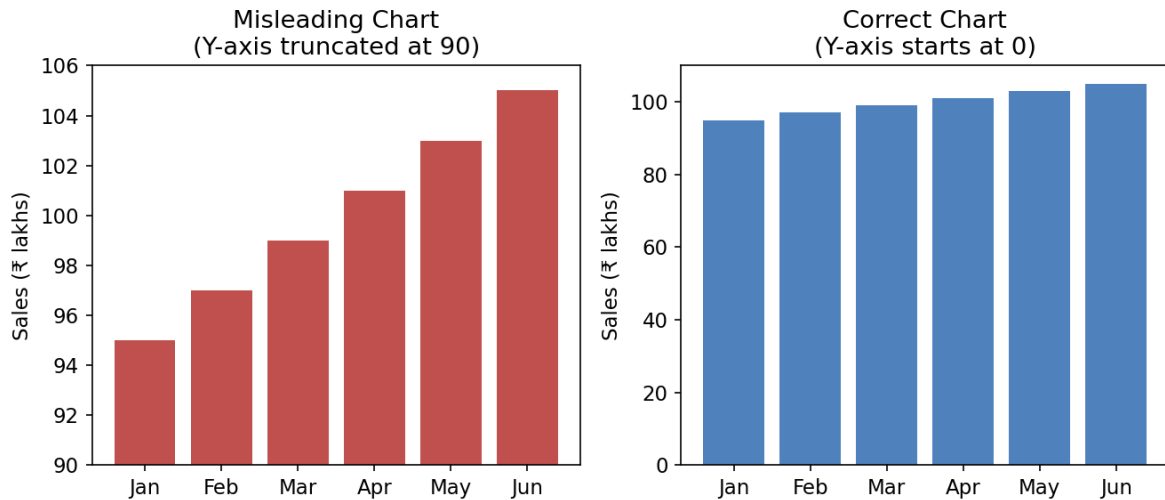
The actual sales grew from 95 to 105 lakhs, an increase of only about 10.5%. However, because the Y-axis starts at 90 rather than 0, the visible bar heights are compressed into a very narrow range near the top. This makes the June bar look nearly twice as tall as the January bar, even though the real difference between them is small. The chart exaggerates the rate of growth and creates a false impression of dramatic performance improvement.

2. Visualization principle violated

This violates the principle of proportional representation (also called the 'zero baseline' or 'truncated axis' principle) in data visualization. For bar charts, the length/height of a bar is meant to be directly proportional to the value it represents, and this only holds true when the axis starts at zero. Truncating the axis breaks the visual-to-data proportionality, resulting in a distorted, misleading comparison.

3. Recommended improved visualization

The Y-axis should start at 0 so that bar heights are truly proportional to sales values. If the intent is to highlight small month-to-month changes, a line chart (with a clearly labeled, non-truncated axis) or a chart with a clearly marked axis break (using a jagged/broken axis symbol) can be used instead, so viewers are not misled. The comparison below shows the same data plotted with a truncated axis versus a correct zero-based axis.



4. Influence on business decisions

If leadership or investors see the truncated chart, they may conclude that the company is experiencing rapid, near-exponential sales growth. This could lead to overly optimistic decisions such as increasing production capacity, hiring aggressively, raising investor expectations, or setting unrealistic future sales targets. When actual performance later fails to match the perceived 'doubling' trend, it could damage credibility with stakeholders and lead to poor resource allocation based on a false premise.

Question 2: Pie Chart with Incorrect Proportions

A university reports student enrollment across four departments:

Department	Students
Computer Science	420
Electronics	280
Mechanical	200
Civil	100

The designer's pie chart shows slices of approximately 50%, 30%, 15%, and 15%, which do not match the true proportions of the data.

1. Correct percentage for each department

Total students = $420 + 280 + 200 + 100 = 1000$.

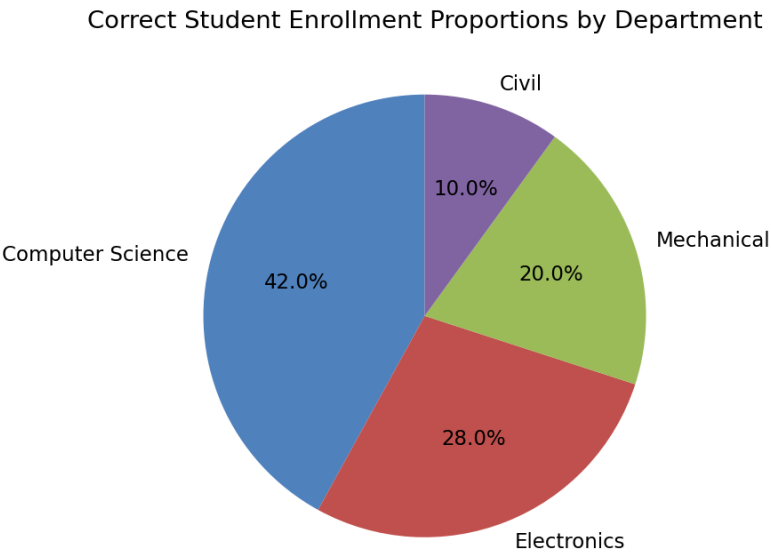
Department	Students	Correct Percentage
Computer Science	420	42.0%
Electronics	280	28.0%
Mechanical	200	20.0%
Civil	100	10.0%

2. Why the displayed chart is misleading

The correct proportions are 42%, 28%, 20%, and 10%, but the chart displays roughly 50%, 30%, 15%, and 15%. This overstates Computer Science and Electronics while understating Mechanical, and it makes Civil (10%) appear equal to Mechanical (20%) at 15% each. Such distortion misrepresents the relative size of each department, making viewers draw incorrect conclusions about which departments are largest or smallest.

3. Alternative chart for improved comparison accuracy

A horizontal or vertical bar chart is generally more accurate than a pie chart for comparing four categories, since the human eye judges bar length more precisely than pie slice angle/area. If a pie chart is preferred for showing 'share of whole,' it must be drawn with exact, data-driven proportions (as shown below) and ideally include percentage labels.



4. Effect on institutional planning

If department heads and administrators rely on the distorted chart, they may over-allocate faculty, budget, lab space, or infrastructure to Computer Science and Electronics while under-resourcing Mechanical, and may fail to recognize that Civil has only half the enrollment of Mechanical. This can result in misaligned staffing ratios, unequal facility investment, and flawed long-term academic planning decisions such as intake quotas or new program approvals.

Question 3: Unequal Bar Widths in Revenue Comparison

A software company's annual revenue by product:

Product	Revenue (₹ Crores)
A	60
B	75
C	90
D	105

The designer used increasing bar widths (in addition to heights) so that Product D appears nearly three times larger than Product A, though its revenue is only 1.75 times that of Product A.

1. Graphical error identified

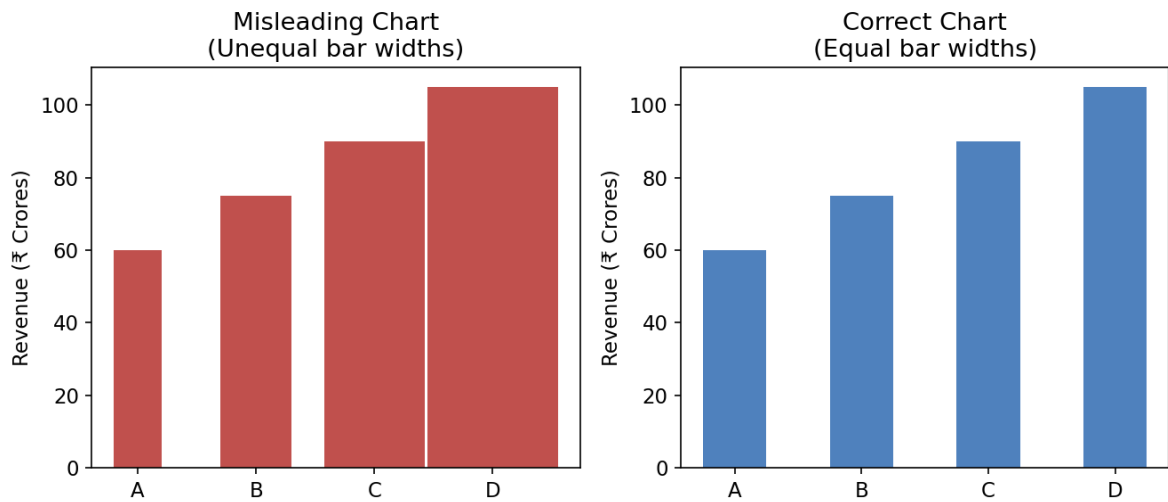
The chart uses unequal bar widths, which is a fundamental graphical error. In a proper bar chart, only the height (or length) of a bar should encode the data value; width must remain constant across all bars. Here, width is also being varied along with height, so the visual area of each bar grows faster than the underlying data.

2. How unequal bar widths distort perception

When both height and width increase together, the perceived area of a bar increases roughly as the square of the linear scale factor. Since Product D's revenue (105) is only 1.75 times Product A's revenue (60), a chart that also widens Product D's bar makes its total visual area appear roughly three times larger, exaggerating the true difference and misleading viewers into overestimating Product D's dominance.

3. Redesign using correct design principles

The corrected design principle is: keep bar width constant for every category, and let only the bar height/length represent the data value, on an axis that starts at zero. Equal spacing between bars should also be maintained for consistent visual comparison.



4. Consequences if investors rely on this chart

Investors viewing the distorted chart might overestimate Product D's contribution to overall revenue and could misjudge the company's growth trajectory or portfolio balance. This could lead to overinvestment in Product D at the expense of other products, mispriced valuations, or misplaced confidence in the company's financial health — decisions that may not hold up once the true, modest differences in revenue become apparent.

Question 4: 3D Column Chart Distortion

An energy company's renewable energy generation data:

Source	Energy (GWh)
Solar	450
Wind	500
Hydro	550
Biomass	520

Because of the 3D perspective and viewing angle used, the Hydro bar appears about 40% larger than Wind, even though the actual difference between them is only 10%.

1. Why 3D charts can create misleading proportions

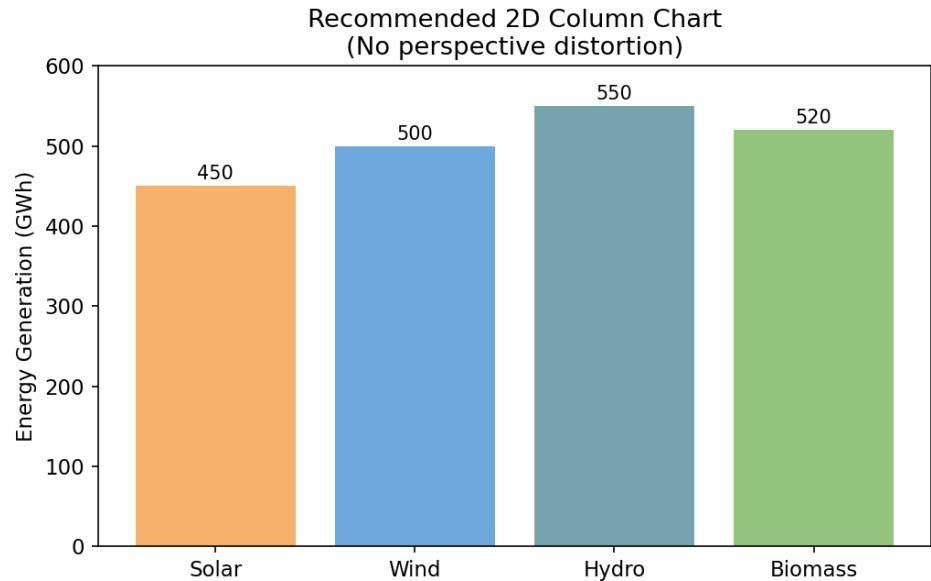
3D charts introduce perspective and depth effects: bars positioned further back or at certain angles can appear larger or smaller than bars of the same actual height positioned differently, purely due to the camera angle and vanishing-point perspective. Viewers instinctively judge apparent size rather than the underlying numeric value, so the added dimension distorts otherwise accurate data without adding any real informational value.

2. Perceived difference vs. actual difference

The actual gap between Wind (500 GWh) and Hydro (550 GWh) is only 50 GWh, or about 10%. However, the 3D perspective effect makes Hydro appear roughly 40% larger than Wind — four times the true difference — purely because of the visual distortion introduced by the 3D rendering and viewing angle, not because of any real difference in the underlying data.

3. Recommended better visualization

A simple flat 2D column (bar) chart, viewed straight-on with no perspective tilt, accurately represents each value through bar height alone. This removes any angle-dependent distortion and lets viewers compare values reliably.



4. Importance of avoiding visual distortion in sustainability reporting

Sustainability and energy reports are often used by regulators, investors, and the public to assess a company's environmental performance and progress toward renewable energy targets. Distorted visuals can create a false impression of which energy source is contributing most, potentially leading to misguided policy decisions, misallocated subsidies or investment, or accusations of 'greenwashing' if the distortion is later discovered. Accurate, undistorted charts are essential for maintaining transparency and stakeholder trust.

Question 5: Bubble Chart with Incorrect Scaling

A healthcare organization's disease case counts by district:

District	Cases
A	150
B	250
C	350
D	450
E	550

The analyst scaled the bubble radius directly in proportion to case counts, making District E's bubble appear more than ten times larger than District A's, even though the actual case count is less than 4 times higher (550 vs 150).

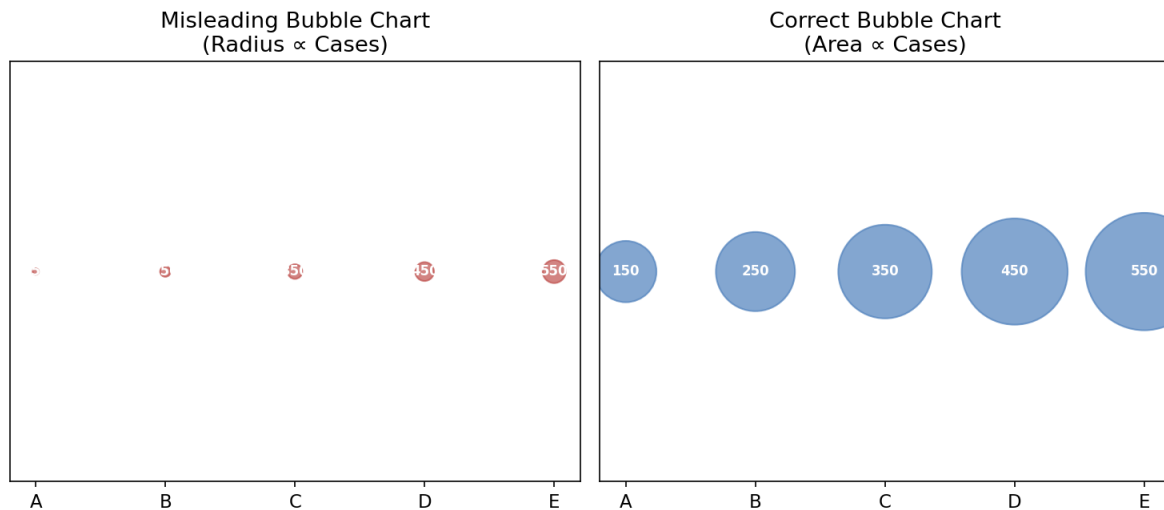
1. Why scaling the radius instead of the area is misleading

The visual impression of a bubble's 'size' is driven by its area, not its radius, because that is what the eye perceives and compares. If radius is made directly proportional to the data value, then area — proportional to radius squared —

grows quadratically. So a district with roughly 3.7 times more cases (550 vs 150) ends up with a bubble whose visible area is about 13-14 times larger, massively overstating the true difference.

2. Correct method for proportional bubble sizing

The area of each bubble should be made directly proportional to the data value, not the radius. Since $\text{area} = \pi \times r^2$, this means the radius must be set proportional to the square root of the data value ($r \propto \sqrt{\text{value}}$). This way, a value that is twice as large produces a bubble with twice the visual area, matching viewer perception to the actual data.



3. Alternative visualization suitable for this dataset

Since this dataset only compares a single quantity (case count) across five districts, a simple bar chart or a choropleth (shaded) map of the districts would communicate the differences more accurately and clearly than a bubble chart. A correctly area-scaled bubble chart (as shown above) is also acceptable when a map-like spatial layout is important.

4. Impact on public health resource allocation

If health officials rely on the radius-scaled bubble chart, they may perceive District E's outbreak as dramatically worse than it actually is relative to other districts, leading to a disproportionate allocation of medical staff, funding, vaccines, or emergency resources toward District E while under-resourcing districts like C and D, whose case counts are also significant. Accurate proportional visualization is critical in public health contexts, where resource-allocation decisions can directly affect patient outcomes.